Applications

1

a 13%

b $\$56\,500$

2

a $\frac{4}{13}$

b $\$1650$

c Option A

3

a $\frac{11}{100}$

b $\frac{6}{55}$

c

x	12	15	25
$P(X=x)$	0.2	0.35	0.45

d $\$18.90$

e $\$5.62$

4

a $\frac{27}{64}$

b $\frac{1}{64}$

c $\frac{27}{64}$

d

x	0	1	2	3
$P(X=x)$	$\frac{27}{64}$	$\frac{27}{64}$	$\frac{9}{64}$	$\frac{1}{64}$

e Yes, because $\Sigma P(x)=1, X$ is discrete numerical, and $0 \leq P(x) \leq 1$.

5

a

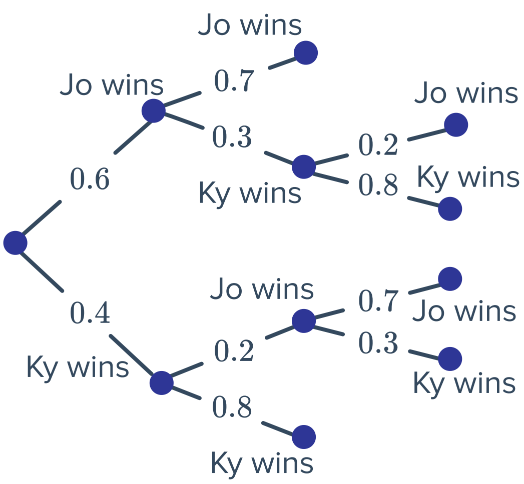
x	0	1	2	3	4
$P(X=x)$	0.4	0.1	0.2	0.2	0.1

b 1.5

c $\frac{1}{5}$

6

a



b

x	2	3
$P(X = x)$	0.74	0.26

c 2.26

d $\frac{23}{128}$

7

a

x	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

b Uniform

c 3.5

d $\frac{35}{12}$

8

a

y	2	3	4	5	6	7	8	9	10	11	12
$P(Y = y)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

b 7

c The distribution is symmetrical.

d 7

e $\frac{35}{6}$

9

a

x	1	2	3	4	5	6
$P(X=x)$	$\frac{2}{9}$	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{1}{9}$

b $\frac{10}{3}$

c $\frac{26}{9}$

10

a

x	2	3	4	5	6
$P(X=x)$	$\frac{1}{6}$	$\frac{5}{24}$	$\frac{5}{24}$	$\frac{5}{24}$	$\frac{5}{24}$

b $\frac{49}{12}$

c $\frac{275}{144}$

11

a

x	1	2	3	4	5	6
$P(X=x)$	0.1	0.1	0.1	0.3	0.1	0.3

b

i 0.2 **ii** $\frac{1}{7}$ **iii** $\frac{1}{3}$ **iv** $\frac{2}{9}$

12

a

y	2	3
$P(Y=y)$	$\frac{1}{2}$	$\frac{1}{2}$

b Uniform

13

a

w	15	16	17	18	19	20
$P(W=w)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

b Uniform

14

a 0, 4, 8

b

x	0	4	8
$P(X=x)$	$\frac{1}{6}$	$\frac{4}{6}$	$\frac{1}{6}$

15

a

	1	2	3	4	5	6
1	0	1	2	3	4	5
2	1	0	1	2	3	4
3	2	1	0	1	2	3
4	3	2	1	0	1	2
5	4	3	2	1	0	1
6	5	4	3	2	1	0

b

x	0	1	2	3	4	5
$P(X=x)$	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{8}{36}$	$\frac{6}{36}$	$\frac{4}{36}$	$\frac{2}{36}$

c $\frac{2}{3}$

d $\frac{9}{10}$

16

a

	1	2	3	4	5	6
1	0	1	2	3	4	5
2	1	0	1	2	3	4
3	2	1	0	1	2	3
4	3	2	1	0	1	2
5	4	3	2	1	0	1
6	5	4	3	2	1	0

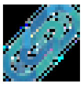
b

x	\$0	\$1	\$2	\$3
$P(X=x)$	$\frac{24}{36}$	$\frac{6}{36}$	$\frac{4}{36}$	$\frac{2}{36}$

c \$0.56

d $-\$1.44$

17



a

x	\$3	\$6	\$9
$P(X=x)$	$\frac{3}{6}$	$\frac{2}{6}$	$\frac{1}{6}$

b \$5

c \$2.24

18

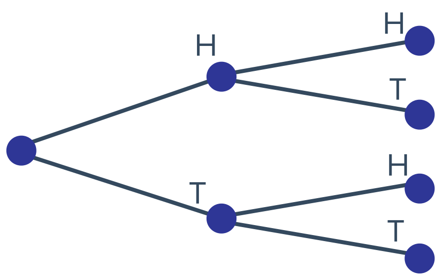
a 0.76 **b** \$1.69 **c** $C = \$1.50$

19

x	0	1	2
$P(X=x)$	0.09	0.42	0.49

20

a



b

x	$-\$3$	\$0	\$3
$P(X=x$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

21

a 0.147

b 0.1029

c 0.07203

d $P(N=n) = (0.7)^n \times 0.3$ for $n = 0, 1, 2, 3, \ldots$

22

a 1, 2, 3, 4, 5, 6, 7, 8

b

x	1	2	3	4	[illegible]	6	7	8
$P(X=x)$	$\frac{3}{16}$	$\frac{3}{16}$	$\frac{2}{16}$	$\frac{2}{16}$	$\frac{2}{16}$	$\frac{2}{16}$	$\frac{1}{16}$	$\frac{1}{16}$

23

a

x	1	2	3	4
$P(X=x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$

b

x	1	2	3	4
$P(X=x)$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{7}{8}$	1

c

i $\frac{3}{4}$ **ii** $\frac{1}{4}$ **iii** $\frac{5}{8}$ **iv** $\frac{3}{4}$

24

a 0, 1, 2, 3, 4, 6, 8, 9, 12, 16

b

x	0	1	2	3	4	6	8	9	12	16
$P(X=x)$	$\frac{9}{25}$	$\frac{1}{25}$	$\frac{2}{25}$	$\frac{2}{25}$	$\frac{3}{25}$	$\frac{2}{25}$	$\frac{2}{25}$	$\frac{1}{25}$	$\frac{2}{25}$	$\frac{1}{25}$

c 4

d 20

25

a

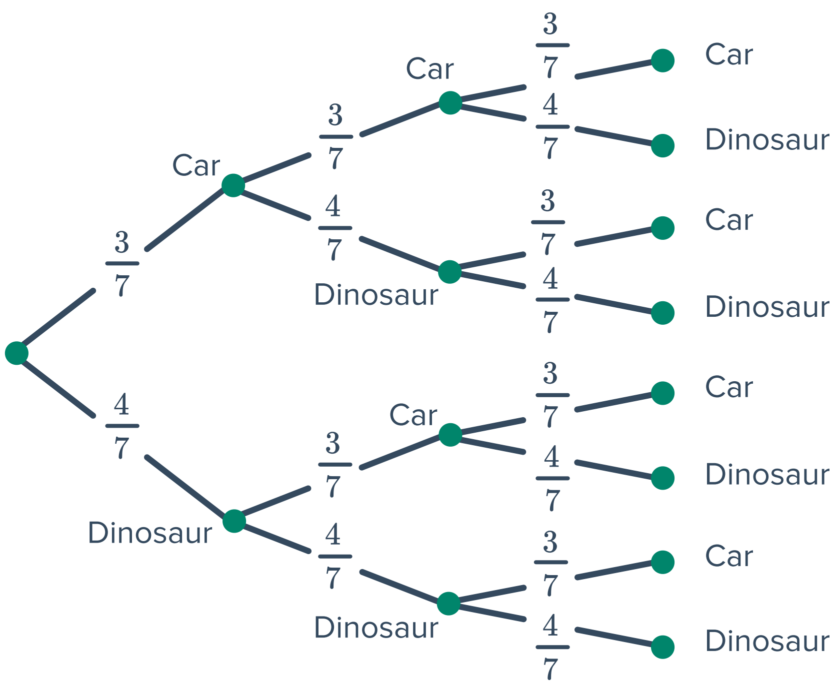
x	0	1	2	3
$P(X=x)$	$\frac{27}{1000}$	$\frac{189}{1000}$	$\frac{441}{1000}$	$\frac{343}{1000}$

b

x	0	1	2	3
$P(X=x)$	$\frac{6}{720}$	$\frac{126}{720}$	$\frac{378}{720}$	$\frac{210}{720}$

26

a



b

x	0	1	2	3
$P(X=x)$	$\frac{64}{343}$	$\frac{144}{343}$	$\frac{108}{343}$	$\frac{27}{343}$

c $1\frac{2}{7}$ days

d $\frac{6}{7}$ days

27

a $\frac{2}{11}$

b $\frac{10}{33}$

c

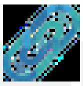
X	0	1	2	3	4
$P(X=x)$	$\frac{1}{66}$	$\frac{2}{11}$	$\frac{5}{11}$	$\frac{10}{33}$	$\frac{1}{22}$

28

a $\frac{1125}{9604}$

b $\frac{3645}{9604}$

c



x	0	1	2	3	4
$P(X=x)$	$\frac{625}{38\,416}$	$\frac{1125}{9604}$	$\frac{6075}{19\,208}$	$\frac{3645}{9604}$	$\frac{6561}{38\,416}$

29

a

x	0	1	2
$P(X=x)$	$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$

b $\frac{1}{6}$

30

a

x	0	1	2
$P(X=x)$	$\frac{7}{15}$	$\frac{7}{15}$	$\frac{1}{15}$

b

t	3	7	11
$P(T=t)$	$\frac{7}{15}$	$\frac{7}{15}$	$\frac{1}{15}$

c $3, 7$

d 5.4